

Passive WiFi Sensing with Motion–Communication Separation for ISAC

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Abstract—WiFi channel state information (CSI) enables low-cost wireless sensing, but passive CSI sensing in integrated sensing and communication (ISAC) settings remains challenging because the CSI stream is shaped not only by human motion but also by communication behavior. Packet arrivals become irregular, packet sources vary over time, and uplink/downlink traffic patterns introduce structured variation that can interfere with activity recognition. In this paper, we study human activity recognition from passively sniffed WiFi communication packets under diverse traffic scenarios, including browsing, YouTube, downloading, uploading, mixed traffic, and near-idle conditions. We propose DART, a dual-branch adversarially routed Time-Decay Gated Recurrent Unit (TDGRU) framework that separates motion-related and communication-related information during representation learning. DART routes motion-dominant and traffic-dominant subcarriers to different encoders, models source-aware timing and PHY metadata in a dedicated traffic branch, and uses adversarial scenario suppression together with orthogonality regularization to reduce communication leakage into the motion representation. Experimental results show that DART achieves 90.60% overall accuracy in in-domain scenario evaluation and 90.54% mean accuracy under leave-one-scenario-out evaluation, outperforming the most relevant baselines in the more communication-sensitive scenarios. Additional experiments with different devices and locations further show promising robustness and reveal the importance of packet direction for future communication-aware sensing design.

Index Terms—ISAC, CSI, WiFi, Passive Sensing.

I. INTRODUCTION

WiFi channel state information (CSI) has become a useful signal for wireless sensing because it captures fine-grained changes in the propagation environment. Prior work has shown that CSI can support human activity and gesture recognition, respiration monitoring, pose-related sensing, and other physical-world inference tasks [1]–[5].

At the same time, integrated sensing and communication (ISAC) is receiving growing attention as wireless systems move beyond communication-only designs. Recent efforts such as *IEEE 802.11bf* highlight the importance of embedding sensing capability into future WiFi systems [6], [7]. In such settings, the same wireless infrastructure is expected to support both communication and sensing [8]. While this integration improves spectrum efficiency and reduces hardware cost, it also introduces new challenges for CSI-based sensing.

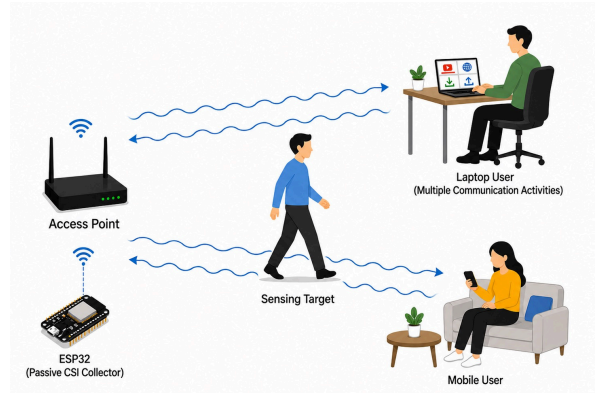


Fig. 1: Human activity recognition through passively sniffed WiFi communication packets.

In passive CSI sensing, the receiver does not control packet transmission and must rely on ambient communication traffic (Fig. 1). As a result, the CSI stream is no longer generated under a stable sampling process. Instead, packet arrivals depend on traffic workload, packet source, and communication dynamics, making the observed CSI sparse, irregular in time, and strongly influenced by communication behavior [9], [10]. Consequently, the sensing signal reflects not only human motion but also when packets arrive, which device transmits them, and whether traffic is dominated by uplink (UL) or downlink (DL) activity. This creates a strong coupling between motion-related and communication-related effects. If not modeled carefully, a sensing system may rely on communication-specific shortcuts rather than learning robust activity patterns.

Recent work has started to address sensing under communication-driven CSI. SenCom [9] demonstrates that sensing can be performed using general WiFi communication packets through CSI extraction and calibration, while UniFi [10] moves further toward fully passive sensing by learning directly from irregularly sampled CSI without packet injection. These approaches make important progress in utilizing communication packets for sensing and handling irregular or heterogeneous CSI streams. However, they do not explicitly incorporate communication behavior, such as traffic type, packet source, and transmission direction, into the sensing model. Practical communication scenarios such as video streaming, downloading, uploading, browsing,

mixed traffic, and near-idle periods produce distinct communication patterns that introduce structured variations in CSI unrelated to human activity. Moreover, uplink traffic, which is left as a future direction in prior work [10], further alters timing and statistical characteristics, highlighting an important gap in communication-aware sensing.

In this paper, we study communication-aware passive CSI sensing under diverse traffic scenarios. Our goal is to learn activity representations that remain robust under irregular packet arrivals while reducing sensitivity to communication-induced variation. To this end, we propose DART, a **D**ual-branch **A**dversarially **R**outed **T**ime-Decay Gated Recurrent Unit (TDGRU) framework, that separates motion-related and communication-related information during representation learning.

The design is motivated by two observations. First, different subcarriers exhibit distinct sensitivities to motion and communication dynamics. Some subcarriers are more informative for human motion, while others are more influenced by communication patterns. Second, packet source and transmission direction introduce different timing characteristics and traffic signatures that can affect CSI. Based on these observations, DART routes motion-dominant and traffic-dominant subcarriers into separate encoders. The motion branch focuses on activity discrimination from irregular CSI, while the traffic branch explicitly models communication behavior using routed subcarriers, source-aware timing cues, PHY metadata, and traffic statistics. In addition, DART incorporates adversarial scenario suppression and orthogonality regularization to further reduce communication-related leakage into the motion representation, while TDGRU enables adaptive modeling of irregular packet arrivals.

The contributions of this study are summarized as follows.

- We identify communication behavior (traffic type, packet source, and direction) as a key confounding factor in passive CSI sensing and systematically study its impact across diverse scenarios.
- We show that subcarriers exhibit different sensitivities to motion and communication dynamics, and leverage this observation through input-level routing into motion and traffic branches.
- We propose DART, a dual-branch framework that reduces coupling between motion-related and communication-related information through subcarrier routing, communication-aware temporal modeling, and adversarial regularization.
- We demonstrate robustness under varying communication conditions, including unseen traffic scenarios, device changes, and location shifts, and highlight packet direction as an important design factor for communication-aware sensing.

The rest of the paper is structured as follows. In Section II, we provide an overview of related studies in the literature. In Section III, we present the details of the proposed DART framework. We then provide our experimental evaluation in

Section IV. Finally, we conclude the study and discuss future directions in Section V.

II. RELATED WORK

A. Active vs Passive WiFi CSI Sensing

WiFi CSI has been widely used for human activity recognition by leveraging fine-grained channel variations caused by human motion [11]–[14]. Recent developments also emphasize the need for standardizing WiFi sensing capabilities, where CSI-based sensing is expected to become a built-in feature of future WiFi systems rather than an ad-hoc research solution [6]. Many existing works follow an *active sensing* paradigm, where CSI is collected through controlled transmitter–receiver communication [15]–[18].

In contrast, *passive sensing*¹ relies on observing (i.e., sniffing) existing WiFi communication signals without requiring dedicated transmitter–receiver coordination for sensing. However, not all passive CSI approaches operate under truly uncontrolled communication conditions. While several studies adopt packet sniffing for CSI collection, many still rely on artificially generated traffic (e.g., high-rate ping or controlled transmissions [22]) to ensure sufficient sampling. More recent efforts [9], [10] move toward true passive sensing using communication packets. Despite this progress, existing approaches do not explicitly model the impact of communication behavior on CSI-based sensing.

Prior active sensing works assume controlled packet generation, while passive sensing works focus on extracting motion-related features from ambient signals. In contrast, our work studies passive CSI under ongoing communication traffic and explicitly models communication-induced variations that can interfere with activity recognition.

B. Integrated Sensing and Communication

Integrated Sensing and Communication (ISAC) has emerged as a key paradigm for next-generation wireless systems, where sensing and communication functionalities are jointly supported using shared spectrum and hardware resources [8]. Recent standardization efforts such as IEEE 802.11bf further promote passive sensing by reusing existing WiFi signaling (e.g., beacons and training frames) without dedicated sensing transmissions [6], [7]. Prior work has explored ISAC from system and architectural perspectives, including joint sensing-communication design, channel estimation, and CSI-aware sensing using advanced signal processing and learning techniques [23]–[27]. These studies highlight the potential of leveraging communication signals for sensing but primarily focus on system-level integration and signal processing aspects.

More recent work has begun to explore CSI-based sensing directly from WiFi communication traffic. SenCom

¹The term “passive” is used broadly in literature, sometimes referring to sensing from ambient RF signals [19]–[21]. Here, we use it specifically for CSI sensing based on packet-level observations without coordinated transmission.

TABLE I: Comparison of communication-driven CSI sensing methods.

	SenCom [9]	UniFi [10]	DART
CSI from real user traffic	✓	✓	✓
No packet injection	×	✓	✓
Multiple packet types	×	✓	✓
Irregular CSI modeling	×	✓	✓
Traffic-type awareness	×	×	✓
Source awareness (UL/DL)	×	×	✓
Motion/traffic separation	×	×	✓
Subcarrier level routing	×	×	✓

[9] demonstrates that CSI can be extracted from general WiFi packets and processed through calibration and fitting techniques to enable sensing without dedicated transmissions. UniFi [10] further advances this direction by learning from irregularly sampled CSI collected from heterogeneous communication packets without packet injection. Related work also shows that CSI sensing is sensitive to sampling conditions: WiImg addresses low packet-rate CSI [28], while [29] mitigates sampling irregularity in WiFi CSI. However, these works do not explicitly model communication-driven traffic variation as a confounding factor.

However, existing works primarily focus on how to utilize communication packets for sensing, without explicitly modeling how communication behavior itself affects CSI. In contrast, our work studies communication-driven variations in CSI and treats communication behavior as an explicit factor in the sensing pipeline. We observe that communication patterns introduce structured variations that can interfere with activity recognition, and design a framework that separates motion-related and communication-related information. This shifts the focus from *using communication packets for sensing* to *modeling and mitigating communication-induced effects* for robust passive CSI-based sensing, as summarized in Table I.

III. PROPOSED DART FRAMEWORK

A. Data Preparation & Input Construction

Fig. 2 illustrates the overall pipeline of the proposed system. After passive CSI data collection under different communication scenarios and human activities, we first preprocess the packet-level CSI data. Specifically, CSI amplitudes are extracted from raw packets and denoised using Hampel filtering followed by moving-average smoothing. Each packet is then represented by its CSI amplitudes, timestamp, source metadata, and PHY metadata. From this packet-level representation, we construct the inputs required by the proposed framework. These include the motion-subcarrier sequence, traffic-subcarrier sequence, inter-arrival sequence, source-feature sequence, PHY-feature sequence, and traffic-statistics vector. The resulting routed inputs are then passed to DART for joint activity recognition and communication-aware representation learning.

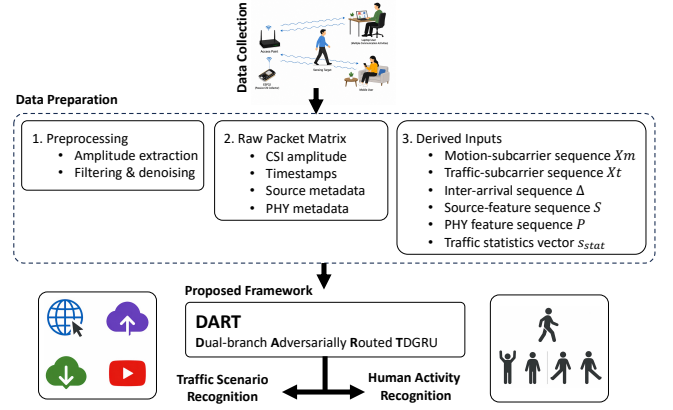


Fig. 2: Overview of the proposed system.

TABLE II: Main notations used in DART framework.

Notation	Description
X_m, X_t	Motion and traffic subcarrier sequences for one window.
τ_i, Δ_i, Δ	Packet timestamp, inter-arrival time, and inter-arrival sequence.
S, P, s_{stat}	Source-feature sequence, PHY-feature sequence, and traffic-statistics vector.
$a_{n,k}, \bar{a}_{n,k}$	Raw and normalized amplitude of subcarrier k for packet n .
$\mathcal{R}$	Reference packet set used for session level normalization.
μ_k, σ_k	Mean and standard deviation of subcarrier k .
$f_{tdgru}^{(m)}, f_{tdgru}^{(t)}$	Motion and traffic TDGRU encoders.
H_m, H_t	Motion and traffic hidden-state sequences.
u_m, u_t	Pooled motion and traffic representations.
$z_{motion}, z_{traffic}$	Motion and traffic embeddings.
$x_i, h_{i-1}, \bar{h}_{i-1}, \gamma_i$	At time i , TDGRU input, previous state, decayed state, and decay factor.
$\hat{y}_{act}, \hat{y}_{scn}, \hat{y}_{adv}$	Activity, scenario, and adversarial scenario predictions.
$GRL(\cdot)$	Gradient reversal layer.
$\mathcal{L}_{act}, \mathcal{L}_{scn}, \mathcal{L}_{adv}, \mathcal{L}_{orth}, \mathcal{L}$	Activity, scenario, adversarial, orthogonality, and total losses.
$\lambda, \lambda_{act}, \lambda_{scn}, \lambda_{adv}, \lambda_{orth}$	GRL strength and loss weights.
L, D_m, D_t	Window length, motion-subcarrier dimension, and traffic-subcarrier dimension.

B. DART Overview

DART is a dual-branch routed TDGRU framework for passive CSI-based human activity recognition under communication-driven sensing conditions. It takes multiple synchronized inputs from each packet window and routes them into two separate encoders. The motion branch learns an embedding z_{motion} for activity recognition, while the traffic branch learns an embedding $z_{traffic}$ for communication scenario modeling. To reduce communication-related leakage in the motion representation, an adversarial scenario head is attached to z_{motion} through a gradient reversal layer. An orthogonality regularizer is also imposed between the two embeddings to encourage stronger separation between

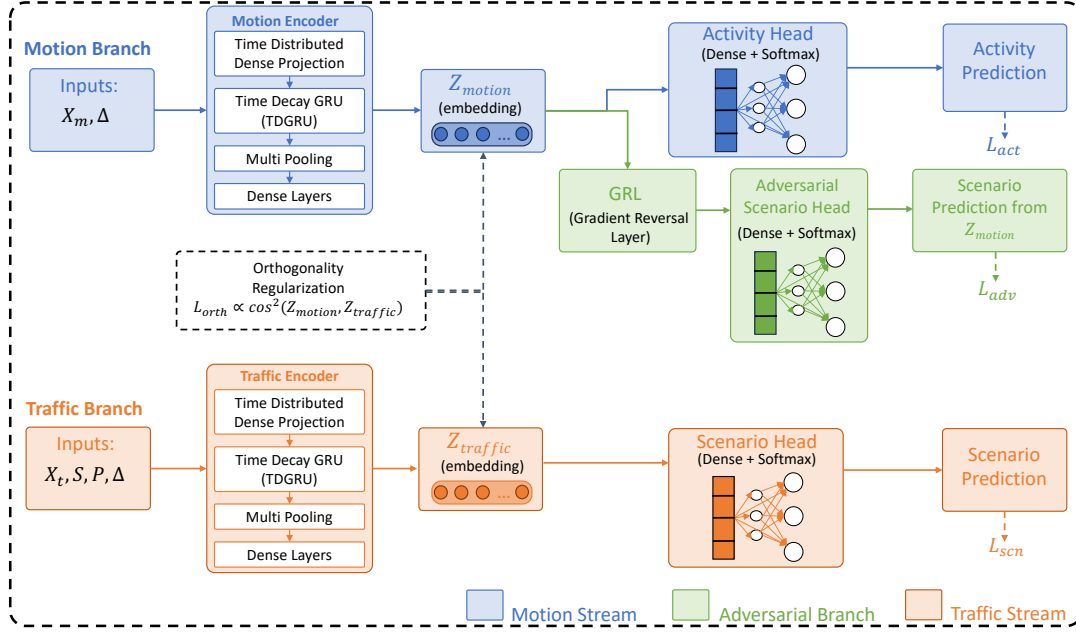


Fig. 3: Architecture of the DART framework.

motion-related and communication-related information. The architecture of DART is shown in Fig. 3 and a summary of the main notations used throughout the paper is provided in Table II.

C. Input Construction and Feature Routing

Before constructing the routed inputs, we normalize CSI amplitudes independently for each session. Let $a_{n,k}$ denote the raw amplitude of subcarrier k for packet n . For each subcarrier, we compute

$$\mu_k = \frac{1}{|\mathcal{R}|} \sum_{n \in \mathcal{R}} a_{n,k}, \quad \sigma_k = \sqrt{\frac{1}{|\mathcal{R}|} \sum_{n \in \mathcal{R}} (a_{n,k} - \mu_k)^2}, \quad (1)$$

and normalize as

$$\bar{a}_{n,k} = \frac{a_{n,k} - \mu_k}{\max(\sigma_k, \epsilon)}, \quad (2)$$

where ϵ ensures numerical stability. This normalization reduces session-dependent amplitude variations and improves generalization across devices and environments.

Let a CSI window contain L packets. After removing null subcarriers, we partition the remaining subcarrier features into motion-dominant and traffic-dominant groups, forming

$$X_m \in \mathbb{R}^{L \times D_m}, \quad X_t \in \mathbb{R}^{L \times D_t}, \quad (3)$$

where D_m and D_t are the numbers of motion and traffic subcarriers, respectively.

Let τ_i denote the timestamp of the i -th packet in the window. We compute the inter-arrival sequence as

$$\Delta_i = \begin{cases} 0, & i = 1, \\ \tau_i - \tau_{i-1}, & i > 1, \end{cases} \quad (4)$$

which gives

$$\Delta = [\Delta_1, \Delta_2, \dots, \Delta_L]^T \in \mathbb{R}^{L \times 1}. \quad (5)$$

For the traffic branch, we construct a source-feature sequence S capturing packet-source indicators and source-dependent timing cues, and a PHY-feature sequence P including physical-layer metadata such as transmission rate and RSSI. In addition, we compute a window-level traffic-statistics vector s_{stat} that summarizes communication patterns within each window, including packet rate, inter-arrival statistics, and source-specific traffic characteristics (e.g., delay).

D. Motion Encoder

The motion encoder takes (X_m, Δ) as input. Each motion-subcarrier vector is first mapped into a learned latent space by a time-distributed linear projection:

$$\tilde{X}_m = \phi_m(X_m). \quad (6)$$

The projected motion sequence is then concatenated with the inter-arrival sequence and passed to a Time-Decay GRU encoder:

$$H_m = f_{\text{tdgru}}^{(m)}([\tilde{X}_m; \Delta]), \quad (7)$$

where $[\cdot; \cdot]$ denotes feature-wise concatenation.

The hidden-state sequence H_m is summarized using attention, average, and max pooling:

$$u_m = [\text{AttnPool}(H_m); \text{AvgPool}(H_m); \text{MaxPool}(H_m)]. \quad (8)$$

The resulting representation is then mapped to the motion embedding:

$$z_{\text{motion}} = g_m(u_m). \quad (9)$$

The embedding z_{motion} captures activity-related motion patterns while suppressing communication-related variations.

E. Traffic Encoder

The traffic encoder follows a similar architecture, but incorporates additional communication-aware inputs. It takes (X_t, S, P, Δ) as input. The traffic-subcarrier sequence is first projected into a learned latent space:

$$\tilde{X}_t = \phi_t(X_t). \quad (10)$$

We then concatenate the projected traffic features with the source features, PHY features, and inter-arrival times:

$$Q_t = [\tilde{X}_t; S; P; \Delta]. \quad (11)$$

This combined sequence is passed to a second TDGRU encoder:

$$H_t = f_{\text{tdgru}}^{(t)}(Q_t). \quad (12)$$

The hidden-state sequence H_t is summarized using the same pooling strategy as in the motion encoder:

$$u_t = [\text{AttnPool}(H_t); \text{AvgPool}(H_t); \text{MaxPool}(H_t)]. \quad (13)$$

The pooled traffic representation is then mapped to the traffic embedding $z_{\text{traffic}} = g_t(u_t)$ to capture communication-scenario characteristics.

F. Prediction Heads

The framework uses three prediction heads. First, the activity head uses the motion embedding to predict the activity label:

$$\hat{y}_{\text{act}} = h_{\text{act}}(z_{\text{motion}}). \quad (14)$$

Second, the communication scenario head uses the traffic embedding together with the traffic-statistics vector:

$$\hat{y}_{\text{scn}} = h_{\text{scn}}([z_{\text{traffic}}; s_{\text{stat}}]). \quad (15)$$

Third, an adversarial scenario head is attached to the motion embedding through a gradient reversal layer:

$$\hat{y}_{\text{adv}} = h_{\text{adv}}(\text{GRL}(z_{\text{motion}})). \quad (16)$$

The first head learns activity discrimination from motion features. The second head models communication scenarios from traffic features. The third head is used to discourage communication scenario information from being encoded in the motion embedding.

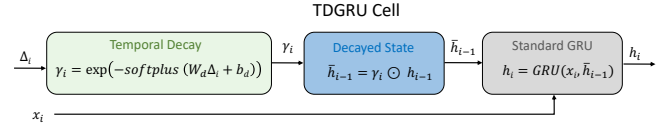


Fig. 4: Structure of the Time-Decay Gated Recurrent Unit (TDGRU)

G. Time-Decay GRU

A standard GRU updates its hidden state using the current input and the previous hidden state. In other words, it learns how much past information should be kept and how much new information should be added at each time step. However, a standard GRU does not explicitly account for irregular time gaps between consecutive packets.

To address this issue, we use a Time-Decay GRU, as illustrated in Fig. 4. At time step i , let x_i denote the current input and let h_{i-1} denote the previous hidden state. We first compute a decay factor from the inter-arrival time Δ_i :

$$\gamma_i = \exp(-\text{softplus}(W_d \Delta_i + b_d)). \quad (17)$$

We then decay the previous hidden state as

$$\bar{h}_{i-1} = \gamma_i \odot h_{i-1}, \quad (18)$$

where $\odot$ denotes element-wise multiplication.

The decayed hidden state is then passed to a standard GRU update:

$$h_i = \text{GRU}(x_i, \bar{h}_{i-1}). \quad (19)$$

This allows the encoder to adapt its memory to packet timing. If the time gap is large, the previous hidden state can be attenuated more strongly. If the time gap is small, more past information can be preserved.

H. Adversarial Scenario Suppression

Although the motion encoder is trained for activity recognition, it may still encode scenario-specific information. To reduce this effect, we use a gradient reversal layer between z_{motion} and the adversarial scenario head. During the forward pass, the gradient reversal layer acts as the identity function. During backpropagation, it multiplies the incoming gradient by $-\lambda$, where $\lambda > 0$ controls the strength of adversarial learning. As a result, the adversarial head tries to predict the scenario label, while the motion encoder is encouraged to make such prediction difficult.

I. Orthogonality Regularization

Even with two separate encoders, the motion and traffic embeddings may still overlap. To reduce this overlap, we impose an orthogonality regularizer. We first normalize the two embeddings:

$$\tilde{z}_{\text{motion}} = \frac{z_{\text{motion}}}{\|z_{\text{motion}}\|_2}, \quad \tilde{z}_{\text{traffic}} = \frac{z_{\text{traffic}}}{\|z_{\text{traffic}}\|_2}. \quad (20)$$

We then define the regularization term as

$$\mathcal{L}_{\text{orth}} = \mathbb{E} \left[\left(\tilde{z}_{\text{motion}}^\top \tilde{z}_{\text{traffic}} \right)^2 \right]. \quad (21)$$

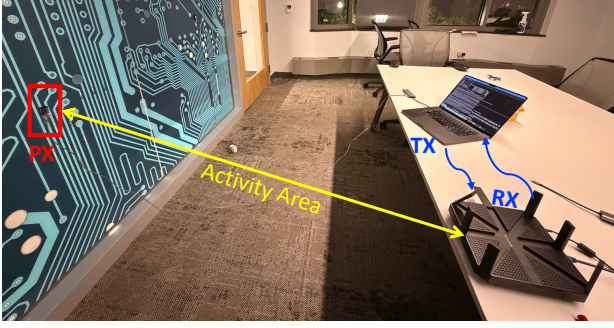


Fig. 5: Experimental setup for data collection.

This term encourages the two embeddings to occupy different directions in the latent space.

J. Training Objective

The full model is trained using four objectives: activity classification, scenario classification from the traffic encoder, adversarial scenario classification from the motion encoder, and orthogonality regularization. The total loss is

$$\mathcal{L} = \lambda_{\text{act}}\mathcal{L}_{\text{act}} + \lambda_{\text{scn}}\mathcal{L}_{\text{scn}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}. \quad (22)$$

Here, $\mathcal{L}_{\text{act}}$, $\mathcal{L}_{\text{scn}}$, and $\mathcal{L}_{\text{adv}}$ are sparse categorical cross-entropy losses, while $\mathcal{L}_{\text{orth}}$ is the orthogonality regularizer.

IV. EVALUATION

A. Dataset

We collected data using an ESP32 device in passive mode with the ESP32 CSI toolkit [30]. Fig. 5 shows the experimental setup. A laptop (TX) communicated with a WiFi router (RX) on channel 3, while the passive ESP32 sniffed CSI from this channel. We followed a round-robin protocol across three human activities (i.e., walking, hand movement (up-down), leg movement (left-right)). For each activity, we recorded 10 repetitions of 10 seconds each. The same activities were repeated under six communication scenarios between the router (RX) and the laptop (TX): browsing, video streaming (YouTube), downloading (a file), uploading (a file), no online activity (i.e., idle), and a mixed scenario combining all of these. To evaluate robustness, we also collected additional data using a smartphone as the transmitter and varying transmitter locations.

B. Communication Characteristics Analysis

Table III provides additional insight into how communication behavior differs across scenarios. The packet rate varies substantially, from only 10.39 packets/s in *Idle* to over 200 packets/s in *Mixed* and *Upload*. The large standard deviations further show that several scenarios are highly bursty rather than temporally stable. This confirms that passive CSI is collected under significantly different communication conditions across scenarios.

The packet source distributions also reveal clear directional differences. *Download* and *YouTube* are more downlink-dominant, with downlink fractions of 68.36% and

TABLE III: Communication characteristics and in-domain HAR accuracy across scenarios.

Scenario	Packet rate (Mean $\pm$ StdDev)	Downlink fraction	Uplink fraction	Accuracy (%)
Overall	—	—	—	90.60
Browsing	93.72 $\pm$ 85.19	51.13	48.87	94.51
Download	138.17 $\pm$ 84.28	68.36	31.64	90.43
Idle	10.39 $\pm$ 19.21	30.96	69.04	92.40
Mixed	224.29 $\pm$ 43.47	53.01	46.99	91.12
Upload	231.09 $\pm$ 24.41	27.33	72.67	89.61
YouTube	55.79 $\pm$ 82.21	60.57	39.43	89.12

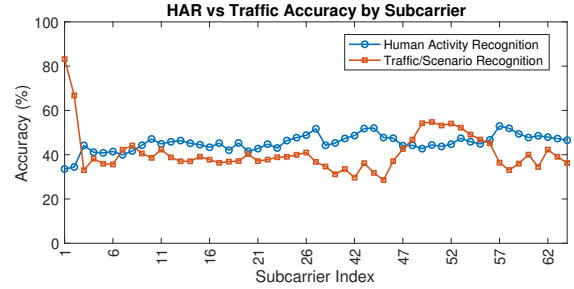


Fig. 6: HAR and communication scenario classification accuracies per-subcarrier.

60.57%, respectively, while *Upload* and *Idle* are more uplink-dominant, with uplink fractions of 72.67% and 69.04%, respectively. In contrast, *Browsing* and *Mixed* are more balanced, with downlink and uplink fractions close to 50%. Together, these results show that each scenario is characterized not only by a different packet rate but also by a different uplink/downlink split. This supports explicit modeling of source-aware traffic behavior rather than treating communication variation as background noise.

C. Subcarrier Routing

We analyze the contribution of individual subcarriers to motion and communication sensing by evaluating classification performance using each subcarrier independently. Fig. 6 shows the per-subcarrier accuracy for HAR and communication scenario classification. We observe that certain subcarriers exhibit stronger sensitivity to communication dynamics than to human motion. In particular, pilot subcarriers (e.g., subcarriers 1 and 2) are highly influenced by communication behavior in passive sensing, as packet arrivals and transmission patterns are not controlled by the receiver. In contrast, these subcarriers tend to remain relatively stable in active sensing setups with controlled transmissions.

More generally, several subcarriers achieve higher accuracy for communication scenario classification than for HAR, indicating a bias toward communication-related variations. Based on this observation, we partition subcarriers into two groups: *traffic-dominant* and *motion-dominant*. The traffic-dominant subcarriers, including the pilot subcarriers, are routed to the traffic branch of the model, while the remaining subcarriers are used for motion encoding. In our

implementation, we select seven traffic-dominant subcarriers based on their relative performance in scenario classification.

D. HAR Evaluation

We evaluate DART under two settings: (i) in-domain evaluation, where training is performed on data from all communication scenarios and testing is conducted on held-out repetitions within each scenario, and (ii) leave-one-scenario-out (LOScO) evaluation, where one scenario is held out for testing and the model is trained on the remaining scenarios. The former evaluates performance when all communication scenarios are observed during training, while the latter evaluates generalization to unseen scenarios.

1) *In-domain evaluation*: Table III reports the overall and the per-scenario activity recognition accuracy on the test set. DART achieves an overall HAR accuracy of 90.60%, indicating that the model can effectively learn activity patterns when all communication conditions are observed during training. The results are consistent across scenarios. The highest accuracy is obtained on *Browsing* at 94.51%, followed by *Idle* at 92.40% and *Mixed* at 91.12%. *Download*, *Upload*, and *YouTube* also achieve strong results, with around 90% accuracy. Overall, these results show that DART performs robustly across different communication scenarios.

2) *Leave-one-scenario-out (LOScO) evaluation*: Fig. 7 reports the activity recognition accuracy for each held-out scenario. DART achieves a mean LOScO accuracy of 90.54%, indicating strong robustness to unseen communication scenarios. The best result is obtained when *Idle* is held out, with an accuracy of 99.22%, followed by *Browsing* at 93.52%. The model also maintains strong performance on *Download*, *Mixed*, and *YouTube*, with accuracies of 90.56%, 90.21%, and 89.75%, respectively. The most challenging case is *Upload*, where the accuracy drops to 80.00%. Overall, these results show that DART generalizes well to unseen communication scenarios, although upload-dominant traffic remains the hardest case.

E. Communication-Only Evidence

To verify that communication scenario is a meaningful source of variation in passive CSI sensing, we evaluate scenario classification using the traffic branch and compare it with the adversarial scenario prediction from the motion branch. The traffic branch achieves 83.75% accuracy for six scenario classes, which shows that the CSI stream contains strong communication-specific patterns. In contrast, the adversarial scenario head produces much lower accuracy (10.05%), close to the six-way chance level of 16.67%. These results confirm that different communication scenarios produce distinct traffic signatures, while the motion branch retains substantially less communication-scenario information than the traffic branch, demonstrating effective separation between motion-related and communication-related representations.

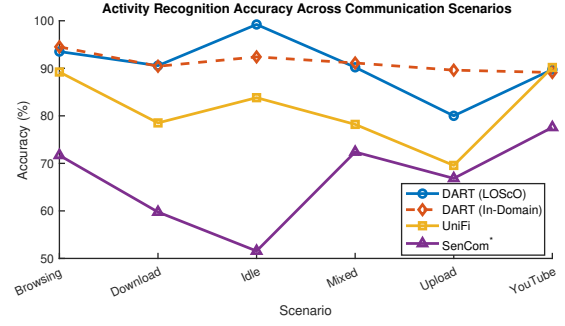


Fig. 7: DART vs. baseline methods under LOScO evaluation. In-domain results are shown for reference.

F. Comparison with Prior Work

We compare DART with two related works, UniFi [10] and SenCom [9]. For UniFi, we implement a baseline following its core idea of directly learning from irregular CSI sequences with timestamp-aware modeling, and evaluate it under the same LOScO setting. For SenCom, since the original study does not provide a complete downstream model for direct reproduction in our setting, we use an injection-style baseline reflecting its communication-assisted sensing approach. As this is not a full implementation of the original framework, we denote it as SenCom*.

Fig. 7 shows that DART performs best in the more communication-sensitive scenarios. Compared with UniFi, DART improves accuracy from 78.51% to 90.56% on *Download*, from 78.20% to 90.21% on *Mixed*, from 69.57% to 80.00% on *Upload*, and from 83.82% to 99.22% on *Idle*. On *Browsing*, DART also improves from 89.22% to 93.52%, while on *YouTube* UniFi is slightly higher at 90.17% compared with 89.75% for DART.

SenCom* performs substantially worse across all scenarios, with accuracies of 77.57% on *YouTube*, 71.69% on *Browsing*, 51.56% on *Idle*, 59.76% on *Download*, 72.39% on *Mixed*, and 66.84% on *Upload*. These results indicate that communication-assisted sensing alone is not sufficient in our setting, where communication behavior must be explicitly separated from motion-related variation.

G. Ablation Study

We examine the contribution of each major design component through a progressive ablation study under the LOScO setting, as shown in Table IV. The variants are arranged from a simpler non-adversarial model to the full DART framework to isolate the effect of each component.

We begin with *No Adv*, which removes adversarial learning and serves as the weakest variant, achieving a mean accuracy of 84.42%. Adding adversarial learning without TDGRU increases the mean accuracy to 86.71%, showing that suppressing scenario information provides a clear benefit even before introducing irregular-time temporal modeling.

Next, we introduce TDGRU without orthogonality regularization, resulting in *Adv + TDGRU (w/o Orth. Reg.)*. This

TABLE IV: LOScO HAR accuracy (%) of DART and its ablated variants across held-out communication scenarios.

Method	Browsing	Download	Idle	Mixed	Upload	YouTube	Mean
No Adv	91.37	85.25	82.33	80.78	73.89	92.87	84.42
Adv w/o TDGRU	85.90	84.71	95.98	86.33	72.41	94.92	86.71
Adv + TDGRU (w/o Orth. Reg.)	93.49	85.59	62.23	87.10	79.81	97.06	84.21
Adv + TDGRU + Orth. Reg. (w/o Routing)	85.96	75.60	100.00	84.17	66.80	93.40	84.32
DART	93.52	90.56	99.22	90.21	80.00	89.75	90.54

TABLE V: Activity recognition accuracy under additional device and location variations.

Evaluation setting	Packet rate (Mean $\pm$ StdDev)	Downlink only (%)	Uplink + Downlink (%)
Different Device	246.96 $\pm$ 74.34	80.03	72.47
Multiple Devices	246.30 $\pm$ 99.11	75.43	69.95
Different Location	235.45 $\pm$ 8.04	70.05	58.97

variant reaches 84.21% mean accuracy. Although it performs strongly on some held-out scenarios such as *YouTube* and *Browsing*, it is less stable overall, especially on *Idle*. This suggests that TDGRU alone is not sufficient without an additional mechanism to better separate motion-related and communication-related information.

When orthogonality regularization is included but routing is still removed, *Adv + TDGRU + Orth. Reg. (w/o Routing)* achieves 84.32% mean accuracy. This indicates that adversarial learning, temporal modeling, and embedding separation are helpful, but without explicit input routing the model still struggles in more communication-sensitive cases such as *Download*, *Mixed*, and *Upload*.

Finally, the full DART model achieves the best result, with a mean LOScO accuracy of 90.54%. The largest gains appear in *Download*, *Mixed*, and *Upload*, with 90.56%, 90.21%, and 80.00%, respectively. Overall, these results show that DART's performance arises from the combined effects of adversarial suppression, irregular-time modeling, orthogonality regularization, and, most importantly, explicit routing between motion-dominant and traffic-dominant inputs.

H. Additional Experiments and Future Directions

To examine deployment-oriented robustness, we collected three additional datasets under the mixed-traffic scenario: i) *Different Device*, where the connected client (TX) changes from laptop to mobile, ii) *Multiple Devices*, where both laptop and mobile are connected simultaneously, and iii) *Different Location*, where the same base setup is used but the laptop is moved to a new position. Table V reports the corresponding activity recognition accuracy using downlink-only CSI and combined uplink-downlink CSI.

These settings exhibit packet-rate variability, indicating stronger unpredictability and communication-side dynamics. Despite this, DART maintains 80.03% accuracy in *Different Device* and 75.43% in *Multiple Devices* using downlink-only CSI, demonstrating robustness under device changes and multi-device traffic.

Across all settings, downlink-only CSI consistently outperforms combined uplink-downlink CSI. This is most ev-

ident in *Different Location*, where accuracy improves from 58.97% to 70.05%. A likely explanation is that the fixed router provides a more stable reference path, while uplink traffic introduces additional variability due to client-side location change. The same tendency is also visible in the *Different Device* and *Multiple Devices* settings, where the mobile device introduces additional communication behavior and burstiness that can make the combined uplink-downlink stream more difficult to model. This highlights that packet direction is an important design factor for robust sensing.

These results also point to future directions. Expanding training data to include diverse device types, multi-device scenarios, and spatial configurations could further improve generalization. More broadly, they motivate communication-aware sensing approaches that account for device heterogeneity, packet direction, and multi-user traffic dynamics.

V. CONCLUSION

In this paper, we study passive WiFi CSI sensing for human activity recognition in realistic ISAC settings, where the sensing signal is strongly influenced by communication behavior. Unlike controlled active sensing, passive CSI collected from ongoing traffic is irregular in time and contains variations induced by traffic type, packet source, uplink/downlink balance, and communication dynamics. To address this problem, we propose DART, a dual-branch adversarially routed TDGRU framework that separates motion-related and communication-related information through sub-carrier routing, source-aware traffic modeling, adversarial scenario suppression, and orthogonality regularization.

Experimental results show that DART performs robustly across diverse communication scenarios. It achieves strong performance in both in-domain and leave-one-scenario-out evaluations, consistently outperforming prior baselines, particularly under download, mixed, and upload traffic. Additional experiments under device and location changes suggest that DART retains generalization ability even under practical deployment shifts, and that downlink CSI can provide a more stable sensing reference in several cases.

Overall, this work highlights the importance of explicitly modeling communication behavior in passive CSI sensing, shifting from treating communication effects as noise to leveraging them as structured signals. Future work can further integrate domain generalization techniques [21], [31]–[33] to improve robustness across environments and users, as well as explore adaptive models that account for device heterogeneity [34] and dynamic traffic conditions in real-world deployments.

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